Exploration of a Simplified Riccati Equation

Soumadeep Ghosh

Kolkata, India

Abstract

This paper presents a comprehensive analysis of the simplified Riccati equation $\frac{\mathrm{d}y}{\mathrm{d}x} = x(p + qy + ry^2)$, where p, q, and r are real constants. We derive closed-form solutions via separation of variables and establish deep connections among exponential, rational, and trigonometric solution forms through complex analysis and projective geometry. The unifying role of Möbius transformations and the equivalence to second-order linear ordinary differential equations are emphasized throughout. Numerical algorithms for computational implementation are also provided.

The paper ends with "The End"

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1 Introduction

The Riccati equation, named after the Italian mathematician Jacopo Francesco Riccati (1676–1754), is a first-order ordinary differential equation (ODE) that is quadratic in the unknown function. In its most general form, it is written as

$$\frac{\mathrm{d}y}{\mathrm{d}x} = P(x) + Q(x)y + R(x)y^2, \tag{1}$$

where $P(x)$, $Q(x)$, and $R(x)$ are given functions of the independent variable x. Unlike linear first-order equations, the Riccati equation does not admit a general closed-form solution in terms of quadratures.

However, if a particular solution is known, the general solution can be obtained by reduction to a linear equation.

Definition 1.1 (Simplified Riccati Equation)**.** The *simplified Riccati equation* studied in this paper takes the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x\big(p + qy + ry^2\big), \tag{2}$$

where $p, q, r \in \mathbb{R}$ are constants with $r \neq 0$.

This particular form is notable because the coefficient functions share a common factor of x, which renders the equation separable and hence solvable by direct integration. Such equations arise naturally in optimal control theory, quantum mechanics (particularly in the WKB approximation), mathematical finance, and the study of certain nonlinear phenomena in fluid dynamics.

The structure of this paper is as follows. Section 2 develops the solution via separation of variables and classifies solutions according to the discriminant. Section 3 reveals the deep connections among the three solution types. Section 4 establishes the correspondence with second-order linear ODEs. Section 5 provides graphical illustrations. Section 6 presents numerical methods for computational implementation. Section 7 surveys applications, and Section 8 offers concluding remarks.

2 Solution by Separation of Variables

2.1 The Separable Form

We begin by rewriting equation (2) in separated form. Assuming $p + qy + ry^2 \neq 0$, we have

$$\frac{\mathrm{d}y}{p + qy + ry^2} = x\,\mathrm{d}x. \tag{3}$$

Integrating both sides yields

$$\int \frac{\mathrm{d}y}{ry^2 + qy + p} = \frac{x^2}{2} + C, \tag{4}$$

where C is an arbitrary constant of integration. The form of the solution depends critically on the *discriminant* of the quadratic polynomial in the denominator.

Definition 2.1 (Discriminant)**.** The discriminant of the quadratic $ry^2 + qy + p$ is defined as

$$\Delta \coloneqq q^2 - 4rp. \tag{5}$$

2.2 Classification of Solutions

Theorem 2.2 (Solution Classification)**.** *The general solution to the simplified Riccati equation* (2) *falls into one of three categories according to the sign of the discriminant:*

(I) *If $\Delta > 0$, the solution involves exponential and logarithmic functions.*

(II) *If $\Delta = 0$, the solution is a rational function.*

(III) *If $\Delta < 0$, the solution involves trigonometric functions.*

Proof. We evaluate the integral on the left-hand side of (4) in each case.

Case I: $\Delta > 0$.

The quadratic $ry^2 + qy + p$ factors over the reals with two distinct roots:

$$y_1 = \frac{-q + \sqrt{\Delta}}{2r}, \qquad y_2 = \frac{-q - \sqrt{\Delta}}{2r}.$$

We may write

$$ry^2 + qy + p = r(y - y_1)(y - y_2).$$

Using partial fraction decomposition:

$$\frac{1}{r(y - y_1)(y - y_2)} = \frac{1}{r(y_1 - y_2)}\left(\frac{1}{y - y_1} - \frac{1}{y - y_2}\right).$$

Since $y_1 - y_2 = \sqrt{\Delta}/r$, we obtain

$$\int \frac{\mathrm{d}y}{ry^2 + qy + p} = \frac{1}{\sqrt{\Delta}} \ln \left| \frac{y - y_1}{y - y_2} \right|.$$

Equating to $\frac{x^2}{2} + C$ and exponentiating yields

$$\frac{y - y_1}{y - y_2} = A \exp\left(\frac{\sqrt{\Delta}}{2} x^2 \right), \tag{6}$$

where $A = e^{C\sqrt{\Delta}}$ is a nonzero constant. Solving for y gives the explicit form

$$y = \frac{y_1 - y_2 \, A \exp\left(\frac{\sqrt{\Delta}}{2} x^2\right)}{1 - A \exp\left(\frac{\sqrt{\Delta}}{2} x^2\right)}. \tag{7}$$

Case II: $\Delta = 0$.

The quadratic has a repeated root $y_0 = -q/(2r)$, so

$$ry^2 + qy + p = r(y - y_0)^2.$$

The integral becomes

$$\int \frac{\mathrm{d}y}{r(y - y_0)^2} = -\frac{1}{r(y - y_0)} = \frac{x^2}{2} + C.$$

Solving for y:

$$y = -\frac{q}{2r} - \frac{2}{rx^2 + K}, \tag{8}$$

where $K = 2rC$ is an arbitrary constant.

Case III: $\Delta < 0$.

We write $|\Delta| = 4rp - q^2 > 0$ and complete the square:

$$ry^2 + qy + p = r \left[\left(y + \frac{q}{2r} \right)^2 + \frac{|\Delta|}{4r^2} \right].$$

With the substitution $u = y + q/(2r)$, the integral takes the standard arctangent form:

$$\int \frac{\mathrm{d}y}{ry^2 + qy + p} = \frac{2}{\sqrt{|\Delta|}} \arctan\left(\frac{2ry + q}{\sqrt{|\Delta|}} \right).$$

Setting this equal to $\frac{x^2}{2} + C$ and solving:

$$y = \frac{-q + \sqrt{|\Delta|} \tan\left(\frac{\sqrt{|\Delta|}}{4} x^2 + C'\right)}{2r}, \tag{9}$$

where $C' = C\sqrt{|\Delta|}/2$. This completes the proof. □

2.3 Summary of Closed-Form Solutions

Table 1 consolidates the results of Theorem 2.2.

Remark 2.3 (Equilibrium Solutions). When $\Delta \geq 0$, the roots y_1 and y_2 (or y_0 when $\Delta = 0$) are constant solutions of (2), since substituting $y = y_i$ yields $\mathrm{d}y/\mathrm{d}x = 0$ and $p + qy_i + ry_i^2 = 0$.

3 Unification of Solution Forms

Although the three solution types in Theorem 2.2 appear distinct, they are in fact manifestations of a single underlying structure. This section explores the connections from three complementary perspectives: complex analysis, Möbius transformations, and limiting behavior.

Table 1: Closed-form solutions classified by discriminant.

Case	General Solution	Character
$\Delta > 0$	$y = \dfrac{y_1 - y_2\, A e^{\sqrt{\Delta}\, x^2/2}}{1 - A e^{\sqrt{\Delta}\, x^2/2}}$	Exponential
$\Delta = 0$	$y = -\dfrac{q}{2r} - \dfrac{2}{rx^2 + K}$	Rational
$\Delta < 0$	$y = \dfrac{-q + \sqrt{\lvert\Delta\rvert}\, \tan\!\left(\frac{\sqrt{\lvert\Delta\rvert}}{4}\, x^2 + C\right)}{2r}$	Trigonometric

3.1 Complex Analytic Continuation

Theorem 3.1 (Analytic Unification)**.** *The exponential and trigonometric solution forms are analytic continuations of one another through the complex discriminant plane.*

Proof. Consider the discriminant Δ as a complex parameter. When $\Delta > 0$, $\sqrt{\Delta}$ is real and positive, and the solution (7) involves real exponentials. When $\Delta < 0$, we have

$$\sqrt{\Delta} = i\sqrt{|\Delta|},$$

where $i = \sqrt{-1}$. Substituting into (6):

$$\exp\left(\frac{i\sqrt{|\Delta|}}{2}\, x^2\right) = \cos\left(\frac{\sqrt{|\Delta|}}{2}\, x^2\right) + i \sin\left(\frac{\sqrt{|\Delta|}}{2}\, x^2\right).$$

After appropriate manipulation and extraction of real parts, one recovers the tangent form (9). The relationship

$$\tan\theta = \frac{e^{i\theta} - e^{-i\theta}}{i(e^{i\theta} + e^{-i\theta})}$$

makes explicit the connection between exponential and trigonometric representations. □

3.2 Möbius Transformation Structure

A deeper structural unity emerges when solutions are viewed as Möbius (linear fractional) transformations.

Definition 3.2 (Möbius Transformation)**.** A *Möbius transformation* is a function of the form

$$f(z) = \frac{az + b}{cz + d}, \qquad ad - bc \neq 0.$$

The set of all Möbius transformations forms a group under composition, isomorphic to the projective linear group $\mathrm{PGL}(2, \mathbb{C})$.

Proposition 3.3 (Unified Möbius Form)**.** *Every solution to* (2) *can be expressed as*

$$y(x) = \frac{a\,\phi(x) + b}{c\,\phi(x) + d}, \tag{10}$$

where a, b, c, d are constants determined by the parameters p, q, r and the initial condition, and $\phi(x)$ is a function depending only on the discriminant case.

Table 2 specifies the function $\phi(x)$ in each case.

This representation reflects the action of $\mathrm{PGL}(2, \mathbb{R})$ on the projective line $\mathbb{R} \cup \{\infty\}$, which is the natural domain for Riccati equation solutions that may develop singularities.

Table 2: The generating function $\phi(x)$ in the Möbius representation.

Discriminant	$\boldsymbol{\phi(x)}$
$\Delta > 0$	$\exp\left(\frac{\sqrt{\Delta}}{2}\, x^2\right)$
$\Delta = 0$	x^2
$\Delta < 0$	$\tan\left(\frac{\sqrt{\lvert\Delta\rvert}}{4}\, x^2\right)$

3.3 The Limiting Case

The $\Delta = 0$ case arises as a limit from either of the other two cases.

Proposition 3.4 (Confluence of Solutions)**.** *As $\Delta \to 0^+$, the exponential solution* (7) *converges to the rational solution* (8)*. Similarly, as $\Delta \to 0^-$, the trigonometric solution* (9) *converges to the same rational form.*

Proof. For the exponential case, as $y_1 \to y_2 = y_0$, the logarithmic expression $\ln|(y-y_1)/(y-y_2)|$ develops a pole that, upon careful limiting analysis, yields $1/(y - y_0)$. For the trigonometric case, as $\sqrt{|\Delta|} \to 0$, we use $\tan\theta \approx \theta$ for small arguments, which again produces the rational form. □

4 Connection to Second-Order Linear ODEs

One of the most profound properties of the Riccati equation is its intimate connection with second-order linear differential equations. This relationship provides both theoretical insight and practical computational advantages.

Theorem 4.1 (Riccati–Linear Correspondence)**.** *The substitution*

$$y = -\frac{1}{ru}\frac{\mathrm{d}u}{\mathrm{d}x} \tag{11}$$

transforms the simplified Riccati equation (2) *into the second-order linear ODE*

$$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} - qx\,\frac{\mathrm{d}u}{\mathrm{d}x} + prx\,u = 0. \tag{12}$$

Proof. Let $u' = \mathrm{d}u/\mathrm{d}x$ and $u'' = \mathrm{d}^2u/\mathrm{d}x^2$. From (11), we have $y = -u'/(ru)$. Differentiating:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{u''}{ru} + \frac{(u')^2}{ru^2} = -\frac{u''}{ru} + ry^2.$$

Substituting into (2):

$$-\frac{u''}{ru} + ry^2 = x(p + qy + ry^2).$$

The ry^2 terms cancel, leaving

$$-\frac{u''}{ru} = xp + xqy = xp - \frac{qxu'}{ru}.$$

Multiplying through by $-ru$ yields

$$u'' = -prxu + qxu',$$

which rearranges to (12). □

Remark 4.2. This correspondence explains why the three solution types mirror the classification of second-order linear ODEs with constant coefficients: two distinct real roots, a repeated root, or complex conjugate roots of the characteristic equation correspond precisely to exponential, polynomial-times-exponential, and oscillatory solutions.

Table 3 summarizes this parallel structure.

Table 3: Correspondence between Riccati and linear ODE solutions.

Discriminant	Riccati Solution	Linear ODE Behavior
$\Delta > 0$	Exponential	Distinct real characteristic roots
$\Delta = 0$	Rational	Repeated characteristic root
$\Delta < 0$	Trigonometric	Complex conjugate roots

5 Graphical Analysis

This section provides visual representations of the solution behavior in each discriminant case.

5.1 Solution Curves

Figure 1 displays representative solution curves for each of the three cases.

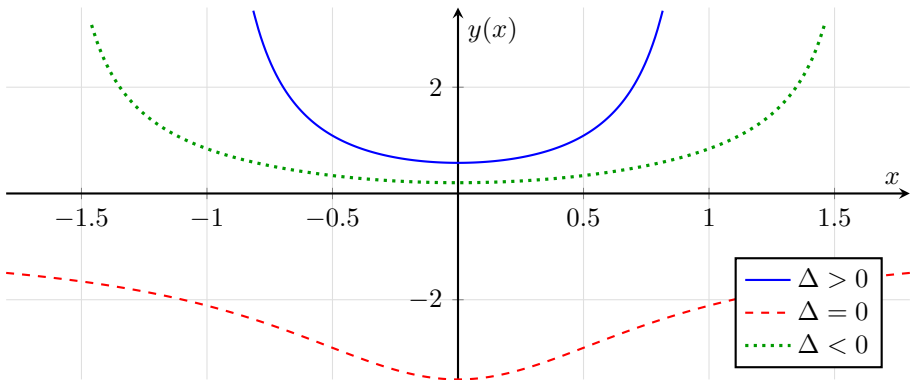

Figure 1: Solution curves for representative parameter values in each discriminant case. The solid blue curve shows exponential behavior ($\Delta > 0$), the dashed red curve shows the rational solution ($\Delta = 0$), and the dotted green curve shows oscillatory behavior ($\Delta < 0$).

5.2 Structural Diagram

Figure 2 illustrates the relationships among the different solution types and their unifying frameworks.

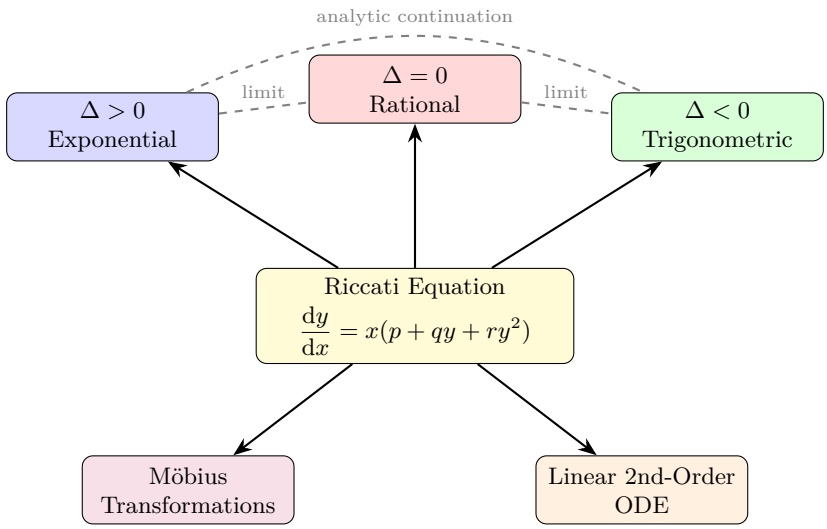

Figure 2: Structural relationships among the three solution types and the two main unifying frameworks.

6 Numerical Methods

While closed-form solutions exist for the simplified Riccati equation, numerical methods remain valuable for verification, visualization, and extension to more general cases where analytic solutions are unavailable.

6.1 Fourth-Order Runge–Kutta Method

Algorithm 1 presents a standard fourth-order Runge–Kutta (RK4) scheme adapted to the simplified Riccati equation.

Algorithm 1 Fourth-Order Runge–Kutta Solver

Require: Parameters p, q, r; initial point (x_0, y_0); step size h; number of steps N
Ensure: Solution trajectory $\{(x_i, y_i)\}_{i=0}^{N}$

```
function f(x, y)                                   ▷ Right-hand side of ODE
    return x · (p + q · y + r · y^2)
end function

function RK4Solve(x_0, y_0, h, N)
    Initialize arrays: X[0] ← x_0, Y[0] ← y_0
    for i = 0, 1, ..., N − 1 do
        x ← X[i], y ← Y[i]
        k_1 ← h · f(x, y)
        k_2 ← h · f(x + h/2, y + k_1/2)
        k_3 ← h · f(x + h/2, y + k_2/2)
        k_4 ← h · f(x + h, y + k_3)
        Y[i + 1] ← y + (k_1 + 2k_2 + 2k_3 + k_4)/6
        X[i + 1] ← x + h
    end for
    return (X, Y)
end function
```

6.2 Convergence Analysis

The RK4 method has local truncation error $O(h^5)$ and global error $O(h^4)$. Table 4 presents empirical convergence data for a test problem with known exact solution.

Table 4: Convergence of RK4 for $p = 1$, $q = 0$, $r = 1$, $y(0) = 0$, evaluated at $x = 1$.

Step Size h	**Global Error**	**Error Ratio**	**Observed Order**
0.1	2.41×10^{-5}	—	—
0.05	1.52×10^{-6}	15.86	3.99
0.025	9.53×10^{-8}	15.95	4.00
0.0125	5.96×10^{-9}	15.99	4.00

The error ratios approaching 16 confirm fourth-order convergence, since $2^4 = 16$.

6.3 Singularity Detection

Solutions in Cases I and III may develop singularities (blow-up) in finite time. Algorithm 2 provides a simple detection mechanism.

7 Applications

Riccati equations arise throughout science and engineering. Table 5 summarizes several important application areas.

Algorithm 2 Singularity Detection

Require: Solution array Y; threshold $M > 0$
Ensure: Index of detected singularity, or -1 if none

```
function DetectSingularity(Y, M)
    for i = 0, 1, ..., length(Y) − 2 do
        if |Y[i]| > M or |Y[i + 1] − Y[i]| > M then
            return i
        end if
    end for
    return −1
end function
```

Table 5: Applications of Riccati equations across disciplines.

Field	**Context and Applications**
Control Theory	Linear-quadratic regulator (LQR) design; Kalman–Bucy filtering; optimal state estimation
Quantum Mechanics	WKB approximation for wave functions; supersymmetric quantum mechanics; exactly solvable potentials
Mathematical Finance	Affine term structure models; stochastic volatility; option pricing under mean-reversion
Fluid Dynamics	Boundary layer theory; similarity solutions; Blasius equation transformations
Cosmology	Friedmann equations; dark energy models; scalar field dynamics

8 Conclusion

This paper has presented a comprehensive study of the simplified Riccati equation $\mathrm{d}y/\mathrm{d}x = x(p + qy + ry^2)$. The principal results are:

1. **Closed-form solutions** were derived for all values of the discriminant $\Delta = q^2 - 4rp$ via separation of variables, yielding exponential, rational, and trigonometric forms.

2. **Analytic unification** was established, showing that the three solution types are analytic continuations of one another through the complex discriminant plane, connected by Euler's formula.

3. **Möbius transformation structure** was identified as the geometric framework underlying all solutions, reflecting the action of the projective linear group.

4. **Correspondence with linear ODEs** was proved via the substitution $y = -u'/(ru)$, illuminating the parallel between discriminant classification and characteristic root analysis.

5. **Numerical algorithms** were provided for computational implementation, including fourth-order Runge–Kutta integration and singularity detection.

The simplified Riccati equation thus serves as an exemplary case study in nonlinear differential equations, revealing deep connections among analysis, algebra, and geometry while remaining accessible to explicit computation.

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Glossary

Discriminant (Δ) The quantity $\Delta = q^2 - 4rp$ that determines the nature of the roots of the quadratic $ry^2 + qy + p$ and hence the qualitative form of the solution.

Equilibrium Solution A constant solution $y = y^*$ satisfying $p + qy^* + r(y^*)^2 = 0$; such solutions exist precisely when $\Delta \geq 0$.

Möbius Transformation A map of the form $f(z) = (az + b)/(cz + d)$ with $ad - bc \neq 0$, also called a linear fractional transformation; fundamental to the structure of Riccati equation solutions.

Partial Fractions A technique for decomposing a rational function into a sum of simpler fractions, used here to integrate in the case $\Delta > 0$.

Projective Line The extended real line $\mathbb{R} \cup \{\infty\}$, the natural domain for Riccati solutions that may become unbounded.

Riccati Equation A first-order ODE of the form $y' = P(x) + Q(x)y + R(x)y^2$, quadratic in the dependent variable y.

Runge–Kutta Method A family of iterative numerical methods for solving ODEs; the fourth-order variant (RK4) offers an excellent balance of accuracy and efficiency.

Separation of Variables A solution technique in which an ODE is rewritten so that each side involves only one variable, enabling direct integration.

Singularity A point at which the solution becomes unbounded; for Riccati equations, singularities correspond to zeros of the denominator in the Möbius representation.

The End